TELIPORT
Script Your Future
Season 4

TATA
TATA ELXSI

Case Study 2

Team Details

THINK. BUILD. CREATE. WIN

TELIPORT
Script Your Future
Season 4

Test a battery before you build it

VOLT-FORGE, a virtual design platform for students and engineers

TATA
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PROBLEM UNDERSTANDING

10 min at 25 W

Keep a Raspberry Pi 5 gateway running through a power cut, inside every cell limit.

DESIRABILITY

No parts needed

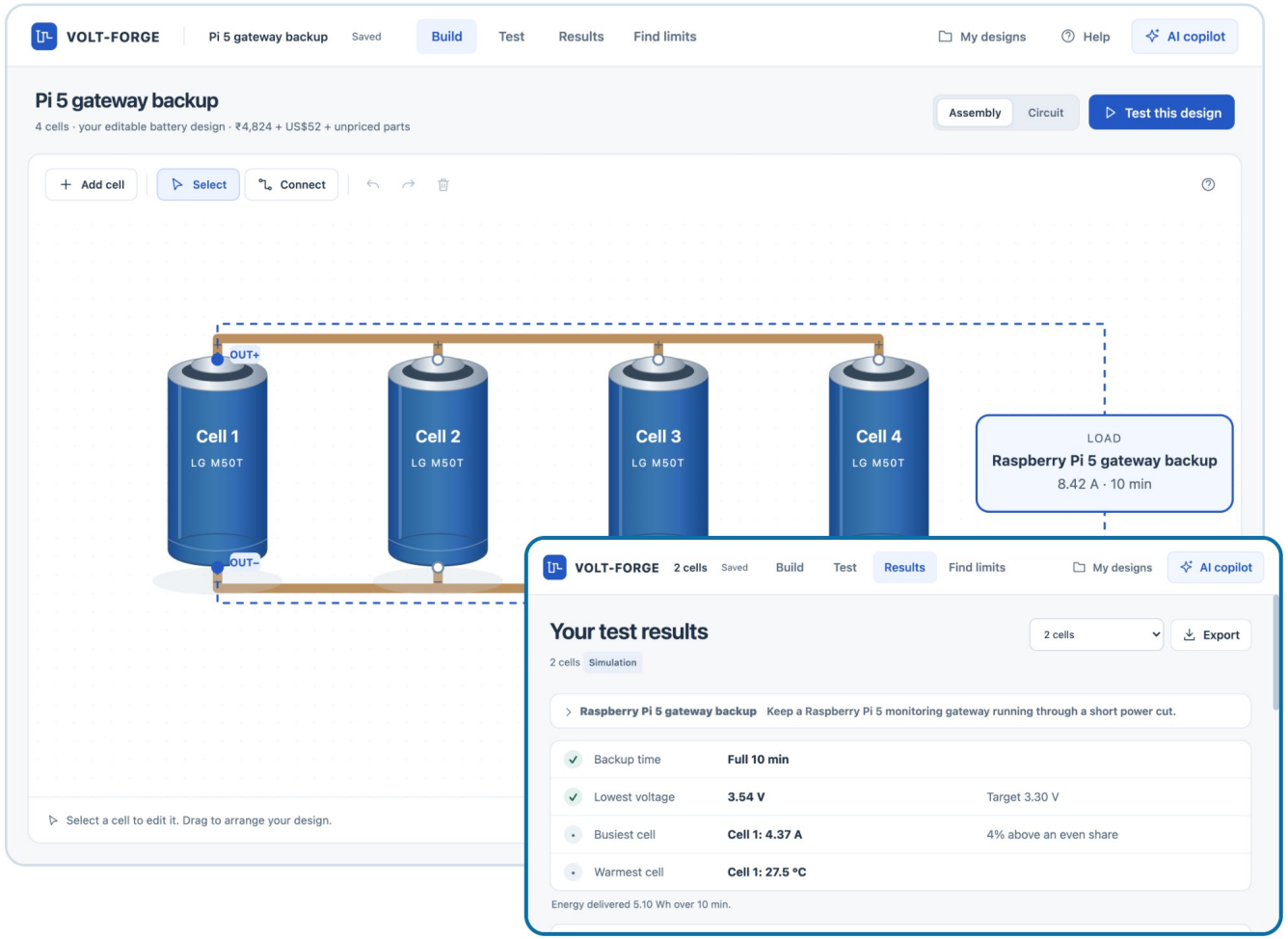
Students and engineers see what a design does before spending on hardware.

TECHNICAL FEASIBILITY

33 of 33

required Pi cases pass with **2 LG M50T cells**. Simple, and within limits.

PROTOTYPE & EVIDENCE



4,006 simulated cases

15 of 15 PyBaMM checks

5 of 5 exports replay exactly

ENGINEERING QUALITY

Energy adds up

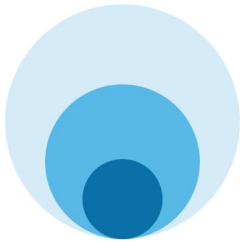
Heat matches electrical loss in every run. Cooling stays a **range**, not a prediction.

INNOVATION

4 → 2 cells

The tool finds the smallest design that still does the job.

BUSINESS VIABILITY



41.3 lakh
TAM: engineering students

15.6 lakh
SAM: electrical, electronics, mechanical

130 colleges
SOM: 5%, assumed

₹2,412 less on cells. Free for students, paid for labs.

Simulated, not yet built. Temperature still differs from real data, so next: bench-test real cells, then user trials.